

Week 1

WEEKLY PROGRESS REPORT

Project: The Oracle That Wears Us

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Report Date: March 09, 2026

Week Period: Mar 02 - Mar 08, 2026

Executive Summary

Week 1 progress: 19 of 19 tasks completed (100%). 5 tasks from future weeks already in progress.

19

COMPLETED

0

INCOMPLETE

5

FUTURE WEEKS

Week Progress

100% complete

2.0

COMPLETED THIS WEEK

These tasks were planned for Week 1 and successfully completed:

✦ HIGHLIGHTS

45 - Unity vs TouchDesigner vs Hybrid: Real-Time Microphone Audio Visualization – Evaluation & Final Technology Choice by berlogabob, naydino, electricianv001

We need to decide once and for all what we will use for the real-time microphone input + visualization pipeline in Project02: - Unity only - TouchDesigner only - Hybrid (Unity + TouchDesigner connected)

5 - TD Test: “Can TD reliably ingest mic audio and turn it into usable control/data?” by berlogabob, naydino

Test to answer the question “How can we take a microphone input and transform it”

20 - Unity: Microphone input test completed by electricianv001

Performance & Evaluation Data Time to setup: 5–10 minutes (including adding components and linking references). Time to response (microphone reaction delay): 1 frame (almost real-time, visually instant). Loading time: Start Unity → 10–15 seconds Open project → 5–10 seconds Scene ready → immediate Stability: The system works consistently without errors once configured correctly. Microphone initialization is stable and does not require additional permissions in this environment. Resource usage: Very lightweight. Uses a small float buffer (256 samples) and simple RMS calculation per frame. Personal evaluation: The solution fits our current project needs well. It provides real-time audio input analysis with smooth visual response. The setup is simple, modular, and easy to integrate into other scenes or objects. At this stage of the project, it is sufficient and flexible for further expansion (e.g., frequency-based analysis if needed).

52 - STT->TTT->TTS Pure Python workflow by berlogabob

Test pure Python workflow for recognise speech to text, connect text to LLM model fow answering in text and give answer as voice back to user.

13 - TTS - Text to Speech KittenTTS by berlogabob

(KittenTTS)(test this model

12 - STT- Speech to Text LLM by berlogabob

(moonshine-a)() test this model. write feedback

6 - TD Test: “Can TD call a free speech-to-text system and return text back into TD as DAT/CHOP?” by berlogabob, naydino, electricianv001

1 - How does the plotter connect to the PC? by berlogabob, naydino

Plotter could be connected to the PC with: 1. USB-cable directly - PROS: Speed, low latency. Robust connectivity. - CONS: wire, visually not so “magic” or “WOW-effect”. 2. Wi-Fi. - PROS: no wires, working like a charm, visually working like “magic” good for our purpose. - CONS: Wi-fi point needed, extra piece of technology. Result: start to work with wire connection, switch to Wi-fi if possible later.

Notes

🌟 Week Summary:

- Engineered a pure Python STT and TTT workflow with integrated LLM capabilities for automated processing.
- Conducted real-time microphone audio testing across Unity, TouchDesigner, and Hybrid environments to verify performance.
- Built project v2 structures for Blanc development, including repository configuration and milestone tracking in Brf systems.
- Designed weekly report generators and auto daily planning tools to streamline administrative workflows.

3.0

NOT COMPLETED THIS WEEK

These tasks were planned for Week 1 but remain incomplete:

All planned tasks completed! 🎉

Blockers & Challenges

Document any blockers or challenges that prevented completion.

4.0

FUTURE WEEK TASKS (STARTED EARLY)

These tasks are scheduled for future weeks but work has already begun:

149 - Future Hardware / IoT Integration Strategy (Week 2)

- how Unity could connect to hardware - possible communication methods - microphone trigger via button - plotter control approach...

148 - System Architecture & Data Flow Definition (Week 2)

- high-level diagram - STT → LLM → Projection → Plotter - I/O boundaries...

58 - Environmental Projection Prototype (Unity) (Week 2)

Focus: - real-time projection - - placeholder data - - - responsiveness - - frame stability...

56 - Oracle output based on user interaction (Week 2)

48 - Test and compare speech to text models inside of TD (Week 2)

test ...

Next Week Preview

Week 2: Mar 09 - Mar 15, 2026

149 - Future Hardware / IoT Integration Strategy

- how Unity could connect to hardware - possible communication methods - microphone trigger via button - plotter control approach...

148 - System Architecture & Data Flow Definition

- high-level diagram - STT → LLM → Projection → Plotter - I/O boundaries...

58 - Environmental Projection Prototype (Unity)

Focus: - real-time projection - - placeholder data - - - responsiveness - - frame stability...

57 - Plotter

Start of conversion laser engraver into 2d graphical plotter...

56 - Oracle output based on user interaction

48 - Test and compare speech to text models inside of TD

test ...

Team

Nadine Allan System Architecture &
Integration

Andrey Dyakov Plotter &
Materialization

Dmitri Kazantsev Generative
Concepts & Fabrication

Risks & Decisions Needed

Document any risks or decisions needed from the team.